

## Chicago Public Library Data Dashboard

Thank you for taking the time to launch my dashboard examining data for Chicago Public Libraries! The following pages will dive deeper into the motivation behind making the dashboard and examine the story behind the data. It will be an analysis of the dashboard itself and an exploration of future improvements. The goal was to turn publicly available but boring CSV files into interesting and accessible visualizations. It is my hope that you find this project successful in that regard!

### AI Statement

Claude was the primary AI tool used for this project. It assisted in creating the `clean_data.py` file from a Jupyter Notebook I had created. This makes the code easier to read and more transferable if more data is added in the future. Claude also assisted in creating the tab format and in debugging the events API. Claude provided most of the code for the directory tab and functionality for clicking the map bubble to display the branch information. It also provided the code for the option to exclude the Harold Washington Library Center.

### The Data

The data comes from the Chicago Data Portal, a public website that hosts many datasets across various domains within the city. The datasets for public libraries are a wealth of knowledge. I elected not to use the holds placed and filled by location datasets because I felt that those numbers were encapsulated within the circulation dataset. I also chose not to use data before 2020. This is because there were many missing data for the variables I wanted to analyze between 2011-2020. I debated using synthetic data to supplement the missing numbers but ultimately decided against it. It would have been very interesting, however, to examine how Covid affected the services provided by the libraries. One could investigate similar data for another large city such as New York City and infer comparable effects.

## The Motivation

I decided to investigate the Chicago Public Library dataset because libraries play an important role within the community, and that can sometimes be taken for granted. I have personally benefited from public libraries, and this project seemed like a positive way to use my growing technical skills to illustrate my gratitude. I wanted to see if I could create visualizations that demonstrate the importance of libraries within communities both physically and digitally.

To achieve this goal, I decided to highlight the total annual visitor count and the digital service ratio. This ratio is calculated by taking the total annual computer sessions per branch and dividing it by the total annual physical (circulations) and digital services rendered per branch. I also wanted to include a map that illustrates how widespread libraries are throughout the city. Not only how many branches there are but that they all are frequented, reflecting their importance in the community. A scatterplot comparing the annual visitors and digital services also makes this comparison more visible.

In addition, I thought it would be relevant to display the most popular branches by year – although, to no surprise, the Harold Washington Library Center dominates this competition. All three visualizations dynamically update when the year is selected, allowing the user to determine what years they want to view.

The events tab was inspired by the desire to display how lively the libraries are. They are not simply static repositories for books. They have an amazing amount of programming every day and for all ages. Similarly, the directory is an attempt to centralize contact information for libraries across the city.

## Analysis

The Harold Washington Library Center is an outlier that skews the data. This is not surprising; it is the most popular for tourists to visit and is a central location downtown. The option to exclude this particular branch is valuable for highlighting the libraries that truly serve the local community beyond providing a respite from tourist activities.

The digital service ratio is the most important visualization in the dashboard. The color changing bubbles make the digital service divide viscerally apparent. This should serve as a launching point to investigate further as to why some communities are so reliant on their library for digital services. It also serves as a heartening reminder that not all branches are solely there for digital services. This suggests that people are still going to the library for physical books.

The short timeline is detriment to the project. It would be fascinating to see how libraries change over the span of 10+ years, particularly during the Covid era.

### Looking forward

This dashboard could be improved with more data or the implementation of synthetic data to enhance the time scale as discussed above. It could also be improved by data displaying the types of resources provided by each library branch, not just the events. This would be another way to illustrate how the library plays a pivotal role in providing essential services as well as targeting what types of resources are important for different communities. The events tab could also be improved for aesthetic appearances with more time.

Given more time and resources, this dashboard could be moved into a permanent website that would support more interaction, including being able to exit out of the branch directory card when interacting with the bubbles on the map.

Thanks again for taking the time to look at this project!